

Lecture: Ordinary Differential Equation

Achraf Badahmane

March 31, 2026



University
Mohammed VI
Polytechnic

THE UM6P
VANGUARD
CENTER

#Empowering Minds.

Plan

- 1 Introduction
- 2 Typical Problems
- 3 The Cauchy Problem

INTRODUCTION

Differential Equations

A differential equation is an equation involving one or more derivatives of an unknown function.

If all derivatives are taken with respect to a single variable, the equation is called an **ordinary differential equation (ODE)**.

An equation involving partial derivatives is called a **partial differential equation (PDE)**.

In this chapter, we consider first-order ordinary differential equations.

TYPICAL PROBLEMS

Section– Typical Problems

Differential equations describe the evolution of many phenomena in various fields, as shown by the following four examples.

Lotka–Volterra Model (Predator–Prey)

- ▶ Prey population: $y_1(t)$

Lotka–Volterra Model (Predator–Prey)

- ▶ Prey population: $y_1(t)$
- ▶ Predator population: $y_2(t)$

Lotka–Volterra Model (Predator–Prey)

- ▶ Prey population: $y_1(t)$
- ▶ Predator population: $y_2(t)$

System of Equations

$$\begin{cases} y_1'(t) = C_1 y_1(t) [1 - b_1 y_1(t) - d_2 y_2(t)], \\ y_2'(t) = -C_2 y_2(t) [1 - d_1 y_1(t)]. \end{cases}$$

Lotka–Volterra Model (Predator–Prey)

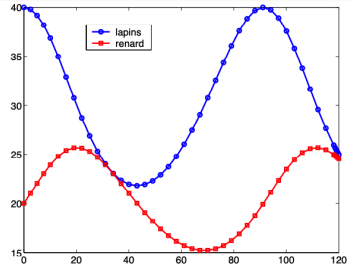
- ▶ Prey population: $y_1(t)$
- ▶ Predator population: $y_2(t)$

System of Equations

$$\begin{cases} y_1'(t) = C_1 y_1(t) [1 - b_1 y_1(t) - d_2 y_2(t)], \\ y_2'(t) = -C_2 y_2(t) [1 - d_1 y_1(t)]. \end{cases}$$

- ▶ C_1 : growth rate of prey
- ▶ C_2 : decay rate of predators
- ▶ d_1, d_2 : interaction coefficients
- ▶ b_1 : food limitation for prey

Predator-Prey Illustration



Paint X Lite

THE CAUCHY PROBLEM

Cauchy Problem (Initial Value Problem)

- ▶ Consider a continuous function $f : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$.

Cauchy Problem (Initial Value Problem)

- ▶ Consider a continuous function $f : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$.
- ▶ Given an initial value $y_0 \in \mathbb{R}$.

Cauchy Problem (Initial Value Problem)

- ▶ Consider a continuous function $f : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$.
- ▶ Given an initial value $y_0 \in \mathbb{R}$.
- ▶ Find a function $y(t)$ defined for $t > 0$ such that:

Cauchy Problem (Initial Value Problem)

- ▶ Consider a continuous function $f : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$.
- ▶ Given an initial value $y_0 \in \mathbb{R}$.
- ▶ Find a function $y(t)$ defined for $t > 0$ such that:

Cauchy Problem

$$\begin{cases} y'(t) = f(t, y(t)), & t > 0, \\ y(0) = y_0. \end{cases}$$

Cauchy Problem (Initial Value Problem)

- ▶ Consider a continuous function $f : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$.
- ▶ Given an initial value $y_0 \in \mathbb{R}$.
- ▶ Find a function $y(t)$ defined for $t > 0$ such that:

Cauchy Problem

$$\begin{cases} y'(t) = f(t, y(t)), & t > 0, \\ y(0) = y_0. \end{cases}$$

$$y'(t) = \frac{dy(t)}{dt}$$

Example: Linear Cauchy Problem

- ▶ A Cauchy problem can be linear.

Example: Linear Cauchy Problem

- ▶ A Cauchy problem can be linear.
- ▶ Consider the function:

$$f(t, v) = 3v - 3t$$

Example: Linear Cauchy Problem

- ▶ A Cauchy problem can be linear.
- ▶ Consider the function:

$$f(t, v) = 3v - 3t$$

- ▶ The associated Cauchy problem is:

Example: Linear Cauchy Problem

- ▶ A Cauchy problem can be linear.
- ▶ Consider the function:

$$f(t, v) = 3v - 3t$$

- ▶ The associated Cauchy problem is:

Linear Initial Value Problem

$$\begin{cases} y'(t) = 3y(t) - 3t, & t > 0, \\ y(0) = 1. \end{cases}$$

Example: Linear Cauchy Problem

- ▶ A Cauchy problem can be linear.
- ▶ Consider the function:

$$f(t, v) = 3v - 3t$$

- ▶ The associated Cauchy problem is:

Linear Initial Value Problem

$$\begin{cases} y'(t) = 3y(t) - 3t, & t > 0, \\ y(0) = 1. \end{cases}$$

Solution

$$y(t) = \left(1 - \frac{1}{3}\right) e^{3t} + t + \frac{1}{3}$$

Cauchy–Lipschitz Theorem

► Let $f : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function.

Cauchy–Lipschitz Theorem

- ▶ Let $f : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function.
- ▶ Assume there exists a constant $L > 0$ such that:

Cauchy–Lipschitz Theorem

- ▶ Let $f : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function.
- ▶ Assume there exists a constant $L > 0$ such that:

Lipschitz Condition

$$|f(t, v) - f(t, w)| \leq L|v - w|, \quad \forall v, w \in \mathbb{R}, \forall t > 0$$

Cauchy–Lipschitz Theorem

- ▶ Let $f : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function.
- ▶ Assume there exists a constant $L > 0$ such that:

Lipschitz Condition

$$|f(t, v) - f(t, w)| \leq L|v - w|, \quad \forall v, w \in \mathbb{R}, \forall t > 0$$

Conclusion

The Cauchy problem admits a **unique global solution** (i.e., defined for all $t > 0$).

Numerical Derivatives with Limits

Let $y : [a, b] \rightarrow \mathbb{R}$ be C^1 , and let $t_0, t_1, \dots, t_n = b$ be $n + 1$ equally spaced nodes in $[a, b]$. We denote $h = \frac{b-a}{n}$ the distance between consecutive nodes. The derivative $y'(t_i)$ is given by:

$$y'(t_i) = \lim_{h \rightarrow 0^+} \frac{y(t_i+h) - y(t_i)}{h}$$

Numerical Derivatives with Limits

Let $y : [a, b] \rightarrow \mathbb{R}$ be C^1 , and let $t_0, t_1, \dots, t_n = b$ be $n + 1$ equally spaced nodes in $[a, b]$. We denote $h = \frac{b-a}{n}$ the distance between consecutive nodes.

The derivative $y'(t_i)$ is given by:

$$y'(t_i) = \lim_{h \rightarrow 0^+} \frac{y(t_i+h) - y(t_i)}{h}$$

$$y'(t_i) = \lim_{h \rightarrow 0^+} \frac{y(t_i) - y(t_i-h)}{h}$$

Numerical Derivatives with Limits

Let $y : [a, b] \rightarrow \mathbb{R}$ be C^1 , and let $t_0, t_1, \dots, t_n = b$ be $n + 1$ equally spaced nodes in $[a, b]$. We denote $h = \frac{b-a}{n}$ the distance between consecutive nodes.

The derivative $y'(t_i)$ is given by:

$$y'(t_i) = \lim_{h \rightarrow 0^+} \frac{y(t_i+h) - y(t_i)}{h}$$

$$y'(t_i) = \lim_{h \rightarrow 0^+} \frac{y(t_i) - y(t_i-h)}{h}$$

$$y'(t_i) = \lim_{h \rightarrow 0^+} \frac{y(t_i+h) - y(t_i-h)}{2h}$$

Finite Difference Approximations of Derivatives

Let $(Dy)_i$ be an approximation of $y'(t_i)$. Then:

$$\text{Forward (Progressive): } (Dy)_i^P = \frac{y(t_{i+1}) - y(t_i)}{h}, \quad i = 0, \dots, n-1$$

Finite Difference Approximations of Derivatives

Let $(Dy)_i$ be an approximation of $y'(t_i)$. Then:

$$\text{Forward (Progressive): } (Dy)_i^P = \frac{y(t_{i+1}) - y(t_i)}{h}, \quad i = 0, \dots, n-1$$

$$\text{Backward (Retrograde): } (Dy)_i^R = \frac{y(t_i) - y(t_{i-1})}{h}, \quad i = 1, \dots, n$$

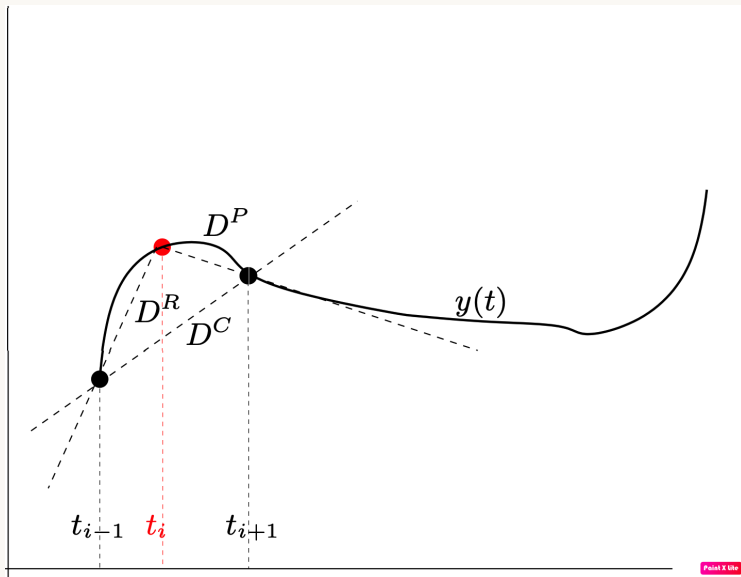
Finite Difference Approximations of Derivatives

Let $(Dy)_i$ be an approximation of $y'(t_i)$. Then:

$$\text{Forward (Progressive): } (Dy)_i^P = \frac{y(t_{i+1}) - y(t_i)}{h}, \quad i = 0, \dots, n-1$$

$$\text{Backward (Retrograde): } (Dy)_i^R = \frac{y(t_i) - y(t_{i-1})}{h}, \quad i = 1, \dots, n$$

$$\text{Central: } (Dy)_i^C = \frac{y(t_{i+1}) - y(t_{i-1}))}{2h}, \quad i = 1, \dots, n-1$$



Finite Difference Methods

Let

$$t_0 = 0 < t_1 < \cdots < t_n < t_{n+1} < \cdots$$

be a sequence of equally spaced real numbers with step size

$$h = t_{n+1} - t_n.$$

Finite Difference Methods

Let

$$t_0 = 0 < t_1 < \cdots < t_n < t_{n+1} < \cdots$$

be a sequence of equally spaced real numbers with step size

$$h = t_{n+1} - t_n.$$

We denote by u_n an approximation of $y(t_n)$.

Finite Difference Methods

Let

$$t_0 = 0 < t_1 < \cdots < t_n < t_{n+1} < \cdots$$

be a sequence of equally spaced real numbers with step size

$$h = t_{n+1} - t_n.$$

We denote by u_n an approximation of $y(t_n)$.

Consider the Cauchy problem:

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0,$$

then at $t = t_n$ we have

$$y'(t_n) = f(t_n, y(t_n)).$$

Finite Difference Methods

Let

$$t_0 = 0 < t_1 < \cdots < t_n < t_{n+1} < \cdots$$

be a sequence of equally spaced real numbers with step size

$$h = t_{n+1} - t_n.$$

We denote by u_n an approximation of $y(t_n)$.

Consider the Cauchy problem:

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0,$$

then at $t = t_n$ we have

$$y'(t_n) = f(t_n, y(t_n)).$$

We also want to approximate the derivative $y'(t_n)$ at the point t_n .

Finite Difference Methods

Let

$$t_0 = 0 < t_1 < \dots < t_n < t_{n+1} < \dots$$

be a sequence of equally spaced real numbers with step size

$$h = t_{n+1} - t_n.$$

We denote by u_n an approximation of $y(t_n)$.

Consider the Cauchy problem:

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0,$$

then at $t = t_n$ we have

$$y'(t_n) = f(t_n, y(t_n)).$$

We also want to approximate the derivative $y'(t_n)$ at the point t_n .

This is done using **numerical differentiation schemes**.

Forward (Progressive) Euler)

$$\frac{u_{n+1} - u_n}{h} = f(t_n, u_n), \quad n = 0, 1, 2, \dots$$

Euler Schemes

Forward (Progressive) Euler)

$$\frac{u_{n+1} - u_n}{h} = f(t_n, u_n), \quad n = 0, 1, 2, \dots$$

Initial condition: $u_0 = y_0$

Euler Schemes

Forward (Progressive) Euler)

$$\frac{u_{n+1} - u_n}{h} = f(t_n, u_n), \quad n = 0, 1, 2, \dots$$

Initial condition: $u_0 = y_0$

Backward (Retrograde) Euler)

$$\frac{u_{n+1} - u_n}{h} = f(t_{n+1}, u_{n+1}), \quad n = 0, 1, 2, \dots$$

Euler Schemes

Forward (Progressive) Euler)

$$\frac{u_{n+1} - u_n}{h} = f(t_n, u_n), \quad n = 0, 1, 2, \dots$$

Initial condition: $u_0 = y_0$

Backward (Retrograde) Euler)

$$\frac{u_{n+1} - u_n}{h} = f(t_{n+1}, u_{n+1}), \quad n = 0, 1, 2, \dots$$

Initial condition: $u_0 = y_0$

Remark 1: Explicit vs Implicit Euler Schemes

Forward (Progressive) Euler is an **explicit** scheme, because it allows to compute

$$u_{n+1} = u_n + hf(t_n, u_n)$$

explicitly from the previous value u_n .

Remark 1: Explicit vs Implicit Euler Schemes

Forward (Progressive) Euler is an **explicit** scheme, because it allows to compute

$$u_{n+1} = u_n + hf(t_n, u_n)$$

explicitly from the previous value u_n .

Backward (Retrograde) Euler is an **implicit** scheme, because

$$u_{n+1} = u_n + hf(t_{n+1}, u_{n+1})$$

is defined **implicitly** in terms of U_n .

Remark 1: Explicit vs Implicit Euler Schemes

Forward (Progressive) Euler is an **explicit** scheme, because it allows to compute

$$u_{n+1} = u_n + hf(t_n, u_n)$$

explicitly from the previous value u_n .

Backward (Retrograde) Euler is an **implicit** scheme, because

$$u_{n+1} = u_n + hf(t_{n+1}, u_{n+1})$$

is defined **implicitly** in terms of U_n .

In general, one must solve a **non-linear equation** at each time step for the **implicit** scheme.

Example 2: Forward Euler in Python

Consider the differential equation with initial condition:

$$\begin{cases} y'(t) = -t y^2(t), & t > 0 \\ y(0) = 2 \end{cases}$$

We want to solve it using Forward Euler for $t \in [0, 4]$, $N_h = 20$, $h = 0.2$.

Example 2: Forward Euler in Python

Consider the differential equation with initial condition:

$$\begin{cases} y'(t) = -t y^2(t), & t > 0 \\ y(0) = 2 \end{cases}$$

We want to solve it using Forward Euler for $t \in [0, 4]$, $N_h = 20$, $h = 0.2$.

The Forward Euler scheme:

$$\begin{cases} \frac{u_{n+1} - u_n}{h} = -t_n u_n^2, & n = 0, 1, 2, \dots, 20 \\ u_0 = 2 \end{cases}$$

Example 2: Forward Euler in Python

Consider the differential equation with initial condition:

$$\begin{cases} y'(t) = -t y^2(t), & t > 0 \\ y(0) = 2 \end{cases}$$

We want to solve it using Forward Euler for $t \in [0, 4]$, $N_h = 20$, $h = 0.2$.

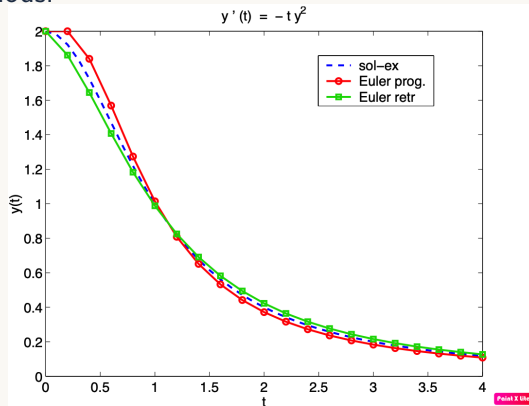
The Forward Euler scheme:

$$\begin{cases} \frac{u_{n+1} - u_n}{h} = -t_n u_n^2, & n = 0, 1, 2, \dots, 20 \\ u_0 = 2 \end{cases}$$

Python implementation:

```
h = 0.2
u = [2]
t = [0, 0.2, 0.4, ..., 4]
for n in range(20): u.append(u[n] + h * (-t[n] * u[n] ** 2))
```


Comparison between the exact solution and the solutions obtained by the Forward and Backward Euler methods.



Second-Order Runge-Kutta Methods

If we integrate the equation

$$y'(t) = f(t, y(t))$$

between t_n and t_{n+1} , we obtain:

$$y(t_{n+1}) - y(t_n) = \int_{t_n}^{t_{n+1}} f(t, y(t)) dt \quad (30)$$

Second-Order Runge-Kutta Methods

If we integrate the equation

$$y'(t) = f(t, y(t))$$

between t_n and t_{n+1} , we obtain:

$$y(t_{n+1}) - y(t_n) = \int_{t_n}^{t_{n+1}} f(t, y(t)) dt \quad (30)$$

Using the trapezoidal rule, we find the following implicit scheme, called the **Crank-Nicolson** or **trapezoidal** scheme:

$$u_{n+1} - u_n = \frac{h}{2} [f(t_n, u_n) + f(t_{n+1}, u_{n+1})], \quad \forall n \geq 0 \quad (31)$$

Second-Order Runge-Kutta Methods

If we integrate the equation

$$y'(t) = f(t, y(t))$$

between t_n and t_{n+1} , we obtain:

$$y(t_{n+1}) - y(t_n) = \int_{t_n}^{t_{n+1}} f(t, y(t)) dt \quad (30)$$

Using the trapezoidal rule, we find the following implicit scheme, called the **Crank-Nicolson** or **trapezoidal** scheme:

$$u_{n+1} - u_n = \frac{h}{2} [f(t_n, u_n) + f(t_{n+1}, u_{n+1})], \quad \forall n \geq 0 \quad (31)$$

This scheme, which is **implicit**, is unconditionally stable when applied to the model problem (16).

Second-Order Runge-Kutta Methods

By modifying scheme (31) to make it **explicit**, we identify **Heun's method**:

$$u_{n+1} - u_n = \frac{h}{2} \left[f(t_n, u_n) + f(t_{n+1}, u_n + hf(t_n, u_n)) \right] \quad (32)$$

Second-Order Runge-Kutta Methods

By modifying scheme (31) to make it **explicit**, we identify **Heun's method**:

$$u_{n+1} - u_n = \frac{h}{2} \left[f(t_n, u_n) + f(t_{n+1}, u_n + hf(t_n, u_n)) \right] \quad (32)$$

Both methods (**Crank-Nicolson** and **Heun**) are of **order 2** with respect to h .

Second-Order Runge-Kutta Methods

By modifying scheme (31) to make it **explicit**, we identify **Heun's method**:

$$u_{n+1} - u_n = \frac{h}{2} \left[f(t_n, u_n) + f(t_{n+1}, u_n + hf(t_n, u_n)) \right] \quad (32)$$

Both methods (**Crank-Nicolson** and **Heun**) are of **order 2** with respect to h .

If we use the **rectangle method** in (30), we obtain:

$$u_{n+1} - u_n = hf(t_{n+\frac{1}{2}}, u_{n+\frac{1}{2}}) \quad (33)$$

Modified Euler Method

If we now approximate

$$u_{n+\frac{1}{2}}$$

by

$$u_{n+\frac{1}{2}} = u_n + \frac{h}{2} f(t_n, u_n)$$

Modified Euler Method

If we now approximate

$$u_{n+\frac{1}{2}}$$

by

$$u_{n+\frac{1}{2}} = u_n + \frac{h}{2}f(t_n, u_n)$$

we obtain the Modified Euler method:

$$u_{n+1} - u_n = hf\left(t_{n+\frac{1}{2}}, u_n + \frac{h}{2}f(t_n, u_n)\right) \quad (34)$$

Modified Euler Method

If we now approximate

$$u_{n+\frac{1}{2}}$$

by

$$u_{n+\frac{1}{2}} = u_n + \frac{h}{2} f(t_n, u_n)$$

we obtain the Modified Euler method:

$$u_{n+1} - u_n = h f\left(t_{n+\frac{1}{2}}, u_n + \frac{h}{2} f(t_n, u_n)\right) \quad (34)$$

Heun's method and the Modified Euler method are particular cases in the family of second-order Runge-Kutta methods.

Modified Euler Method

If we now approximate

$$u_{n+\frac{1}{2}}$$

by

$$u_{n+\frac{1}{2}} = u_n + \frac{h}{2} f(t_n, u_n)$$

we obtain the Modified Euler method:

$$u_{n+1} - u_n = h f\left(t_{n+\frac{1}{2}}, u_n + \frac{h}{2} f(t_n, u_n)\right) \quad (34)$$

Heun's method and the Modified Euler method are particular cases in the family of second-order Runge-Kutta methods.

When applied to the model problem (16), in both cases the stability condition is

$$h < \frac{2}{|\lambda|}$$

Summary of Methods

The table below summarizes the characteristics of the methods we have introduced:

Method	Explicit/Implicit	Step	Stability	Order
Forward Euler	Explicit	1	Conditionally	1
Backward Euler	Implicit	1	Unconditionally	1
Crank-Nicolson	Implicit	1	Unconditionally	2
Heun	Explicit	1	Conditionally	2
Modified Euler	Explicit	1	Conditionally	2

Comparison of Solutions for $h = 0.4$

The first figure below shows the solutions obtained by the two methods as well as the exact solution

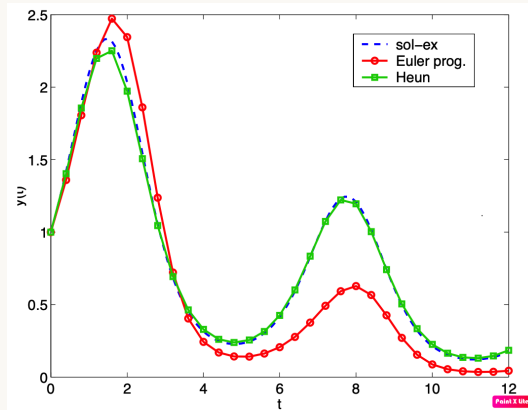
$$y(t) = e^{-0.1t + \sin(t)}.$$

We notice that the solution obtained with **Heun's method** is much more accurate than the one obtained with **Forward Euler**.

Moreover, if we reduce the time step, the solution obtained by Forward Euler approaches the exact solution.

The second figure shows the solutions obtained with $h = 0.4, 0.2, 0.1, 0.05$ using the following commands:

Comparison between Forward Euler and Heun:



Solutions obtained by the Forward Euler method for different time steps:

